

Who Benefits from Federation?

Heterogeneous Gains in Cross-Operator Network Traffic Representation Learning

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Abstract—Research and education networks, regional Internet service providers, and campus networks each hold operational telemetry that would be valuable to model jointly, yet non-disclosure agreements, institutional privacy policy, and commercial sensitivity make centralized pooling infeasible; consequently each operator trains in isolation. We evaluate federated representation learning as a response, constructing a 20-client federation from real telemetry spanning ten Internet2 backbone routers and ten CESNET3 academic institutions, with a commercial regional ISP held out as an unseen-network probe, and comparing centralized, federated, and local-only training at equal optimization budget across five seeds. Federated training proves statistically equivalent to centralized pooling (two one-sided tests, $\pm 5\%$ bound, $p = 0.020$), establishing that the data-custody constraint costs nothing in representation quality. More consequentially, the benefit of federation is not uniform: backbone clients gain a median 9.2% reduction in reconstruction error while campus clients gain 1.9% (Mann-Whitney $p = 0.0003$), an effect that survives outlier removal, is not explained by training-set size, and tracks a difference in traffic regularity we characterize but do not isolate. Two independent manipulations, proximal regularization strength and local-epoch budget, each selectively suppress this benefit for irregular-traffic clients while leaving regular-traffic clients unaffected, identifying unconstrained local optimization as the mechanism and indicating that in this setting client heterogeneity is signal rather than noise.

Index Terms—federated learning, network telemetry, research and education networks, representation learning, client heterogeneity, privacy-preserving collaboration

I. INTRODUCTION

Research and Education Networks (RENs) such as Internet2, ESnet, and GÉANT form the transport substrate for data-intensive science, moving petabyte-scale datasets between instruments, supercomputing facilities, and collaborating institutions [1], [2]. The operational telemetry these networks generate is precisely the raw material needed to build predictive models of network behavior, and precisely the data that cannot be shared: Internet2 flow records are governed by non-disclosure agreements, campus traffic carries institutional privacy obligations, and commercial provider data is competitively sensitive. Models of network behavior are therefore built one operator at a time, with every participant re-deriving what its peers have already learned.

Federated learning permits joint modeling without joint data custody: rather than moving data to the model, it moves the model to the data, exchanging only parameter updates [3]. The paradigm is established in cross-silo settings and has been applied to network traffic classification [4]–[6]. Two questions specific to multi-operator networking nevertheless remain open. The first is how much representation quality is sacrificed by declining to pool data. The second, which to our knowledge has not been posed, is whether participants benefit *equally*, and thus whether an operator considering joining can expect aggregate gains to apply to its own infrastructure.

This paper answers both. We construct a 20-client federation from real telemetry: ten Internet2 backbone routers, dominated by scheduled machine-driven scientific transfers, and ten CESNET3 academic institutions, following human working rhythms. A commercial regional ISP is held out and used at evaluation time as an unseen-network probe. Each client trains an autoencoder over 24-hour traffic windows, an unsupervised representation task rather than classification, and we compare three regimes at equal optimization budget: centralized pooling (an upper bound that violates the data-custody constraint), federated averaging, and local-only training (prevailing practice). Fig. 1 summarizes the design. Our contributions are:

- the first federated representation-learning evaluation spanning a national research backbone, a European National Research and Education Network (NREN), and a commercial regional ISP as heterogeneous clients on real operational telemetry;
- a demonstration that federated training is statistically *equivalent* to centralized pooling ($p = 0.020$, TOST at $\pm 5\%$), rather than merely not significantly different;
- identification of *regime-stratified benefit*: backbone clients gain a median 9.2% against local-only training while campus clients gain 1.9% ($p = 0.0003$), an effect robust to outlier removal and unexplained by training-set size, though the groups differ in more than traffic regime;
- a mechanism established by 2 independent manipulations, proximal strength & local-epoch budget, each selectively suppresses the benefit for irregular-traffic clients; and

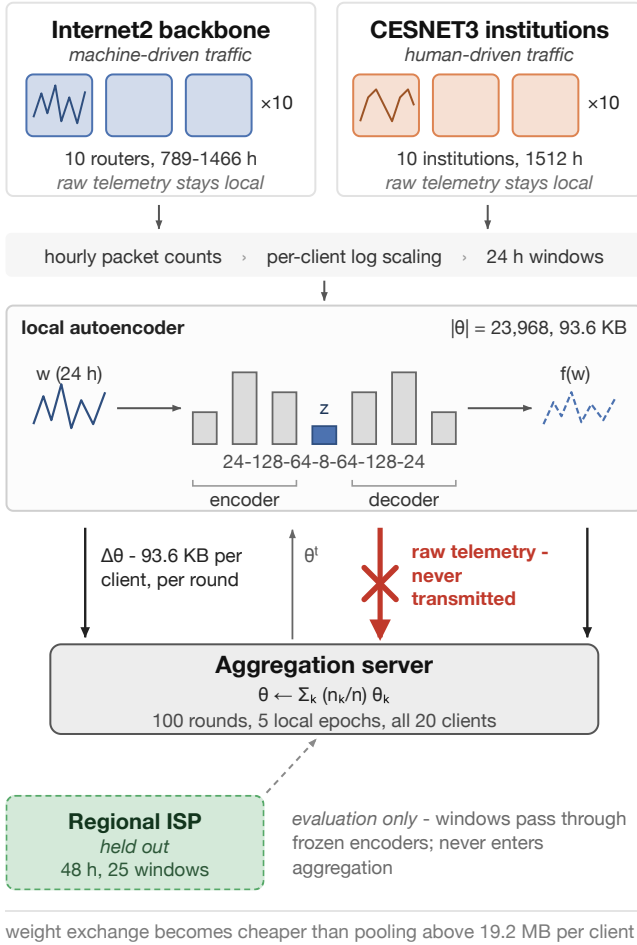


Fig. 1. Federated representation learning across 20 heterogeneous network clients. Only model parameters cross administrative boundaries; raw telemetry never leaves the operator that collected it. A commercial regional ISP is excluded from training and used only as an unseen-network probe.

- operator-facing participation guidance with a communication accounting showing weight exchange becomes bandwidth-competitive only above approximately 19 MB of telemetry per client.

Section II reviews related work, Section III formalizes the federation, Section IV details the setup, Section V presents our results, Section VI discusses their implications and limitations, and Section VII concludes the paper.

II. BACKGROUND AND RELATED WORK

A. Traffic in Research and Education Networks

Research backbones differ structurally from commercial and campus networks. Where consumer-facing networks aggregate many small, human-initiated flows, REN backbones exhibit a heavy-tailed distribution in which a few large scientific transfers carry most of the volume [7]. These transfers are typically scheduled by batch and grid systems rather than initiated interactively, decoupling load from human activity cycles and suppressing diurnal periodicity, an effect reinforced

by the Science DMZ pattern [2], observed likewise at HPC facilities [8], and characterized in detail for the backbone studied here [9]. The difficulty of learning a model of normal behavior from a single vantage point is therefore operator-dependent: a network repeating on a predictable cycle is largely learnable from its own history, whereas one driven by uncoordinated external scheduling is considerably less so.

Predictive modeling of such traffic has advanced steadily, but almost entirely within the boundary of a single operator, whether applying recurrent architectures to backbone forecasting [10], establishing anomaly-aware benchmarks for security baselining [11], or documenting the ingestion machinery required at this volume [12]. Because evaluation is conducted on one network, whether a model or even the relative ordering of competing models transfers to another operator remains untested.

B. Federated Optimization and Client Heterogeneity

Federated averaging [3] established the basic protocol: a server broadcasts a global model, clients perform local epochs, and the server aggregates parameters weighted by local dataset size [13]. Its dominant difficulty is statistical heterogeneity, since local optimization pulls parameters toward local optima, and both empirical characterization [14] and convergence analysis [15] show accuracy degrades as divergence between client distributions grows. This is generally framed as *client drift*, a pathology to be corrected: FedProx [16] penalizes movement from the global model and SCAFFOLD [17] corrects the drift direction via control variates, though no algorithm dominates across non-IID regimes [18]. A related line examines the local-epoch budget, where additional epochs reduce communication rounds but increase local-global divergence [19]. Our federation is heterogeneous in an unusually strong sense, its clients differing by orders of magnitude in volume and belonging to two qualitatively distinct traffic regimes, and we find the standard framing incomplete: the drift these methods suppress proves productive for a specific subset of clients.

A parallel body of work reaches a compatible conclusion from the opposite direction. Rather than forcing agreement, personalized federated learning retains client-specific adaptation, treating the global model as a meta-initialization for local fine-tuning [20], decoupling personalized models through a Moreau envelope [21], or partitioning the network into a shared representation and per-client heads [22]. This last formulation is closest to our results, which suggest empirically that a shared representation combined with unconstrained local adaptation is what benefits irregular-traffic clients. We do not evaluate personalized methods, but Section VI argues our mechanism finding directly motivates them in this domain.

C. Privacy Guarantees and Their Limits

The premise that withholding raw data withholds the information it contains is not exact. Gradient inversion attacks reconstruct training samples from shared updates alone, recovering inputs by optimizing dummy data to match observed gradients [23], with subsequent work improving both fidelity

and practicality [24]. For multi-operator networking this means parameter exchange weakens rather than eliminates disclosure, and a deployment under contractual obligation should treat it as a reduced-exposure channel rather than a closed one. We do not evaluate mitigations such as differentially private update perturbation, and return to this in Section VI.

D. Representation Learning for Network and Time-Series Data

Our objective, learning a compressed description of normal behavior without labels, follows a long lineage from autoencoders as nonlinear dimensionality reduction [25] to the broader programme of learning representations that expose explanatory factors of variation [26]. The appeal here is specific: operational REN telemetry carries no application or attack labels, so a supervised objective presumes annotation deployment cannot supply. Federated learning in networking has nonetheless been applied predominantly to supervised classification, whether approaching centralized accuracy for application identification without transferring packets [4], extending to in-network deployment on constrained hardware [6], or surveyed across the design space [5]; in parallel it has been applied to time-series modeling, where dataset-level heterogeneity across organizations is the central obstacle to shared representations [27], [28].

Two gaps remain: unsupervised representation learning is comparatively unexamined in the multi-operator setting, and evaluations report aggregate performance, which does not serve an operator deciding whether to join a federation whose participants may resemble its own network very little. To our knowledge no prior work asks which participants benefit, or why. Sections V-C and V-D address this directly.

III. FEDERATION DESIGN AND PROBLEM FORMULATION

A. Client Construction

We construct $K = 20$ training clients from two operationally distinct networks, plus a third held out for evaluation (Table I).

Ten clients are Internet2 backbone routers observed over 57 days. We reduce raw NetFlow to hourly packet counts using the out-of-core pipeline of our prior work [12] and use no other field. Coverage ranges from 789 to 1466 hours, reflecting differing collection windows rather than filtering.

Ten further clients are academic institutions from the CESNET-TimeSeries24 corpus [29], a public dataset of per-institution aggregates from the Czech national research and education network. From its 283 institutions we select the ten with highest mean hourly packet volume and truncate each to its first 1512 hours. Truncation is deliberate: retaining the full 40-week corpus would give CESNET3 clients roughly six times the training data of Internet2 clients, confounding any performance difference with data quantity.

The third network is a commercial regional ISP, represented by a 48-hour snapshot. This is too short to support a partition under the eligibility criterion of (4) and is excluded from

TABLE I
COMPOSITION OF THE FEDERATION. WINDOW COUNTS AT $L = 24$.

Network	K	Traffic regime	Hours	Train	Test
Internet2	10	machine-driven	789–1466	529–1003	96–197
CESNET3	10	human-driven	1512	1035	204
Regional ISP [†]	—	broadband	48	—	—
Total	20			18,939	

[†]Held out from training; 25 evaluation windows from 48 hours.

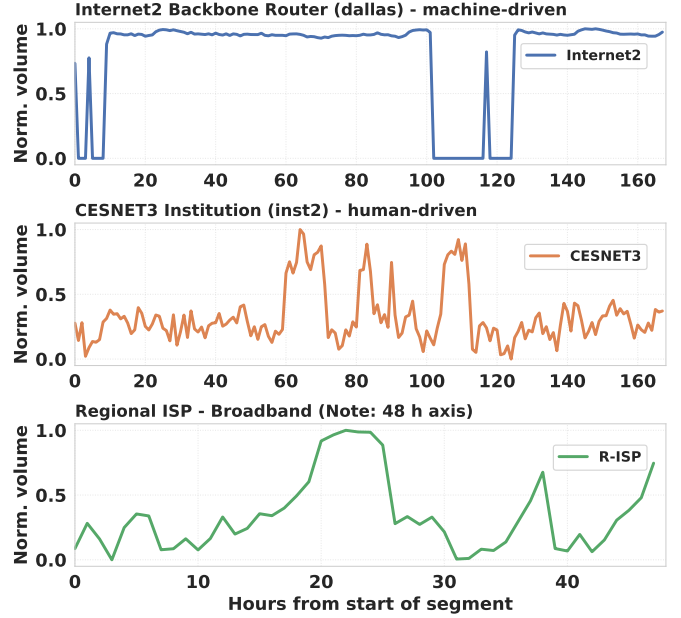


Fig. 2. Three traffic regimes, normalized independently per panel. Top: one week from an Internet2 backbone router, machine-driven. Middle: one week from a CESNET3 institution, human-driven. Bottom: the held-out regional ISP; note this panel spans 48 hours rather than 168. The backbone series contains 31 zero-traffic hours from scheduled maintenance, 29 in its first week; the segment shown begins at hour 241 to avoid them.

training; it is instead reserved as an unseen-network probe, its windows passed through frozen encoders at evaluation time.

The groups differ not merely in scale but in traffic character. Fig. 2 contrasts one week from each. The backbone series shows sustained utilization with abrupt aperiodic transitions and weak daily periodicity; the institutional series shows a pronounced diurnal cycle tracking European working hours with reduced weekend load. We call these the *machine-driven* and *human-driven* regimes. The distinction is the axis along which our principal result is organized.

B. Problem Formulation

Let $\mathcal{C} = \{c_1, \dots, c_K\}$ denote the federation. Client c_k holds a private hourly series $X^{(k)} = (x_1^{(k)}, \dots, x_{T_k}^{(k)})$ with $x_t^{(k)} \in \mathbb{R}_{\geq 0}$ the packet count in hour t . No client observes another's series at any point.

Each series is partitioned chronologically,

$$X^{(k)} = X_{\text{tr}}^{(k)} \oplus X_{\text{va}}^{(k)} \oplus X_{\text{te}}^{(k)}, \quad \rho = (0.70, 0.15, 0.15), \quad (1)$$

since the evaluation question is whether a representation learned on past traffic describes future traffic. Scaling statistics are estimated on the training partition alone:

$$z_t^{(k)} = \frac{\log(1 + x_t^{(k)}) - \mu_k}{\sigma_k}, \quad (\mu_k, \sigma_k) \text{ from } X_{\text{tr}}^{(k)}. \quad (2)$$

Normalization is strictly per-client: a federation-wide scaler would require exchanging distributional statistics, precisely the disclosure the protocol avoids. It also removes absolute magnitude, which spans two orders of magnitude across clients, so that any structure the encoder recovers reflects temporal *shape* rather than volume.

Each partition is independently windowed,

$$\mathcal{D}_s^{(k)} = \left\{ \mathbf{w}_i^{(k)} = (z_i^{(k)}, \dots, z_{i+L-1}^{(k)}) \in \mathbb{R}^L \right\}, \quad (3)$$

with $L = 24$ and unit stride. Windowing within rather than before partitioning is essential: boundary-straddling windows would share 23 of 24 hours across the split.

A client is admitted only if every partition yields at least $N_{\min}$ windows, giving

$$T_k \geq T_{\min} = \left\lceil \frac{N_{\min} + L - 1}{\min(\rho_{\text{va}}, \rho_{\text{te}})} \right\rceil = 354 \text{ hours} \quad (4)$$

for $N_{\min} = 30$, $L = 24$. All twenty training clients satisfy it, including the two shortest at 789 and 861 hours. The regional ISP, at 48 hours, yields zero validation and test windows; its exclusion is arithmetic rather than discretionary.

Each client trains an autoencoder $f_\theta = g_\phi \circ e_\psi$ with $e_\psi : \mathbb{R}^L \rightarrow \mathbb{R}^d$ and $g_\phi : \mathbb{R}^d \rightarrow \mathbb{R}^L$, minimizing

$$\mathcal{L}_k(\theta) = \frac{1}{|\mathcal{D}_{\text{tr}}^{(k)}|} \sum_{\mathbf{w} \in \mathcal{D}_{\text{tr}}^{(k)}} \|\mathbf{w} - f_\theta(\mathbf{w})\|_2^2. \quad (5)$$

The task is unsupervised because operational REN telemetry carries no application or attack labels. The architecture is a symmetric MLP, $24 \rightarrow 128 \rightarrow 64 \rightarrow 8 \rightarrow 64 \rightarrow 128 \rightarrow 24$, with $|\theta| = 23,968$ parameters, sized by data rather than compute: at roughly one thousand windows per client this is about twenty-four parameters per sample, near the point at which local overfitting destabilizes averaging.

C. Training Regimes

We compare three regimes bracketing the operator's design space. Centralized training pools all windows, violating the data-custody constraint and serving as an upper bound:

$$\theta_{\text{cen}}^* = \arg \min_{\theta} \sum_{k=1}^K \frac{n_k}{n} \mathcal{L}_k(\theta), \quad n_k = |\mathcal{D}_{\text{tr}}^{(k)}|, \quad n = \sum_j n_j. \quad (6)$$

Local-only training solves K independent problems, exchanging nothing, and describes prevailing practice:

$$\theta_k^* = \arg \min_{\theta} \mathcal{L}_k(\theta), \quad k = 1, \dots, K. \quad (7)$$

Federated training uses federated averaging with full participation [3]:

$$\theta^{t+1} = \sum_{k=1}^K \frac{n_k}{n} \theta_k^{t+1}, \quad \theta_k^{t+1} \leftarrow \text{LocalOpt}(\theta^t, \mathcal{D}_{\text{tr}}^{(k)}, E). \quad (8)$$

With $K = 20$ we sample no client subset, eliminating participation variance. To test whether heterogeneity-aware aggregation alters our conclusions we also evaluate FedProx [16], which penalizes deviation from the round's global model:

$$\mathcal{L}_k^{\text{prox}}(\theta; \theta^t) = \mathcal{L}_k(\theta) + \frac{\mu}{2} \|\theta - \theta^t\|_2^2, \quad (9)$$

with $\mu = 0$ recovering (8) exactly, which we use as an implementation control.

D. Evaluation Metrics

Representation quality is measured as reconstruction error on each client's held-out windows, referred to throughout as *reconstruction* MAE to avoid ambiguity with forecasting error. Since each client is evaluated under all three regimes, comparisons are paired at the client level. The quantity of central interest is the improvement a client obtains by participating:

$$B_k = 100 \times \frac{\text{MAE}_k^{(\text{loc})} - \text{MAE}_k^{(\text{fed})}}{\text{MAE}_k^{(\text{loc})}} \quad [\%], \quad (10)$$

where $B_k > 0$ indicates a gain. Reporting B_k per client rather than only in aggregate is what permits the question in the title to be answered.

Three statistical procedures are used: a one-sided Wilcoxon signed-rank test for whether federation improves on local-only training across clients; a one-sided Mann-Whitney U test for whether benefit differs between regimes; and, since a non-significant difference does not establish equivalence, the two one-sided tests procedure [30] for federated against centralized. We set the equivalence bound at $\pm 5\%$ of the centralized mean, chosen as an operationally negligible margin: it is smaller than the spread we observe between individual clients and comparable to the between-seed variation of the baselines themselves.

Finally, over R rounds federated averaging transfers

$$C_{\text{fed}} = R \cdot K \cdot 2 \cdot |\theta| \cdot b \quad (11)$$

bytes ($b = 4$; the factor two covers both directions), while centralizing raw telemetry transfers $C_{\text{cen}} = KV$ for per-client volume V . Since the former is constant in V and the latter linear, they are equal at

$$V^* = 2R|\theta|b \approx 19.2 \text{ MB per client}. \quad (12)$$

IV. EXPERIMENTAL SETUP

A. Implementation and Budget Parity

We ran all experiments on a single NVIDIA H200 GPU with 16 CPU cores. The complete federation occupies 2.53 MB in single precision and is resident in device memory throughout, so no host-device transfer occurs within the optimization loop. We implement the models in PyTorch and optimize with Adam [31]; full configuration appears in Table II.

A comparison across regimes is meaningful only if none receives more optimization, so we fix a common budget: federated training performs $R = 100$ rounds of $E = 5$ local epochs, and both baselines are permitted the same 500

TABLE II
MODEL AND TRAINING CONFIGURATION.

Parameter	Value
Architecture	24→128→64→8→64→128→24
Latent dimension d	8
Parameters $ \theta $	23,968 (93.6 KB, float32)
Window length L	24 hours, unit stride
Partition p	0.70 / 0.15 / 0.15, chronological
Optimizer	Adam, lr 10^{-3} , weight decay 10^{-5}
Batch size	64
Rounds R / local epochs E	100 / 5 (budget 500, all regimes)
Client participation	1.0 (all clients, every round)
Early-stop patience	25 (baselines only)
Seeds	42, 43, 44, 45, 46

epochs, though neither exhausts it under early stopping. Model selection nonetheless differs: the baselines use validation early stopping with patience 25, whereas federated averaging retains the round with the lowest mean validation error across clients, since local early stopping is not part of the algorithm. The regimes are therefore matched on budget but not on selection.

B. Reproducibility and Convergence

Every configuration is run under five seeds governing initialization, batch ordering, and the distinct initialization of each client’s model under local-only training. Aggregate results are the mean across seeds of the per-seed cross-client mean, with dispersion across seeds, so that reported variability is run-to-run rather than cross-client.

Because the best global round frequently occurred near the end of the budget, we verified that $R = 100$ does not truncate an improving trajectory. Extending one seed to 200 rounds, mean validation error reached its minimum at round 98 and rose slightly thereafter. The reported configuration operates at the convergence optimum.

V. RESULTS

A. Federation Matches Centralized Pooling

We first establish what the data-custody constraint costs. Federated and centralized training are indistinguishable in reconstruction error (Table III); neither a paired t -test ($p = 0.774$) nor a Wilcoxon signed-rank test ($p = 0.813$) detects a difference across seeds.

Since a non-significant difference is not evidence of equivalence at five paired observations, we test equivalence directly. The two one-sided tests procedure with a $\pm 5\%$ bound rejects a difference larger than that bound in either direction ($p_{\text{TOST}} = 0.020$), establishing statistical equivalence rather than merely undetected difference. The observed difference favors federation, but at 0.46% of the centralized mean against a between-seed deviation of roughly 3.8% it lies well within run-to-run variation and we make no directional claim. We note that the selection asymmetry of Section IV-A can only favor federation, since its checkpoint is chosen against pooled rather than per-client validation data; that federation is

TABLE III
TEST PERFORMANCE BY REGIME. MEAN ACROSS 5 SEEDS OF PER-SEED CROSS-CLIENT MEAN, ± 1 STANDARD DEVIATION ACROSS SEEDS.

Regime	Recon. MSE	Recon. MAE	Probe acc.
Centralized [‡]	0.1532 ± 0.0058	0.2519 ± 0.0064	0.750
Federated	0.1525 ± 0.0033	0.2504 ± 0.0038	0.710
Local-only	0.2618 ± 0.0068	0.3079 ± 0.0024	0.654

[‡]Violates the data-custody constraint; upper bound only. Probe accuracy: logistic regression on frozen embeddings recovering network membership, 70/30 split (Sec. V-F). Significance tests reported in Sec. V-A and B.

nonetheless equivalent to, rather than better than, centralized training suggests the asymmetry does not drive this result.

The practical reading is direct: an operator refusing to export raw telemetry sacrifices nothing measurable in representation quality.

B. Federation Helps, But Not Uniformly

Against local-only training the picture changes: federation reduces mean reconstruction MSE substantially (Table III), confirmed by a one-sided Wilcoxon test ($p = 0.0001$). Counted per client, federation improves reconstruction MAE for 15 of 20 clients by more than two percent, is within two percent for 4, and is worse for 1. Part of this margin reflects model selection rather than representation quality: local-only models early-stop on their own small validation partitions, a choice Section V-F shows overfits, at a validation-to-test ratio of 3.09 against 1.74.

The aggregate requires care. As a mean across clients, local-only training incurs 71.7% higher error; as a median, 4.9%. The discrepancy is almost entirely one client: the most data-constrained Internet2 router, with 529 training windows against a federation median of 1019, improves by 66.7% under federation and is otherwise roughly three times worse than any other client when trained alone. Removing it reduces the mean gap to 10.7%, whereas removing any other single client changes it by at most 13.6 percentage points. We therefore rely on the median wherever a conclusion could otherwise rest on this client.

Fig. 3 shows what the aggregate obscures: the distribution is not merely dispersed but structured.

C. Benefit Is Stratified by Traffic Regime

Grouping B_k by network partitions the distribution cleanly (Fig. 4). Internet2 clients obtain a median benefit of +9.2% (mean +21.6%); CESNET3 clients obtain +1.9% (mean +1.3%). A one-sided Mann-Whitney U test rejects equality ($p = 0.0003$); the twenty clients come from two networks rather than twenty independent operators, so this separates these client sets, not two general populations. Every Internet2 client benefits, with a minimum of +3.6%; the single client federation harms and all four neutral cases are CESNET3 institutions.

The stratification does not depend on the outlier. Excluding the largest beneficiary reduces the Internet2 mean to +16.6%,

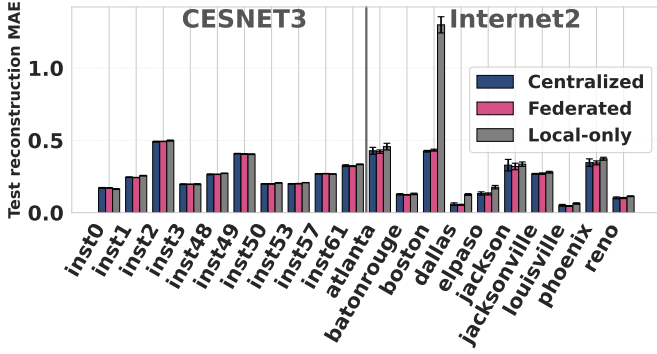


Fig. 3. Test reconstruction MAE per client under all three regimes, mean $\pm$ one standard deviation across five seeds. CESNET3 institutions left of the divider, Internet2 routers right.

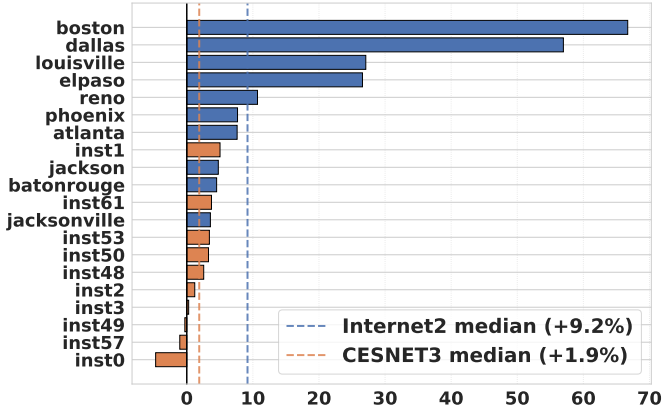


Fig. 4. Per-client federation benefit B_k , sorted, with per-network medians marked. Every backbone client gains; neutral and negative cases are all campus institutions.

and excluding the two largest to +11.6%, both far above the CESNET3 mean; the Mann-Whitney test remains significant without it ($p = 0.0005$), and the median separates the groups by nearly a factor of five by construction.

A natural alternative explanation is data quantity, since Internet2 clients hold 529–1003 training windows while every CESNET3 client holds exactly 1035. Pooled across all clients, benefit and training-set size are indeed correlated ($r = -0.62$, $p = 0.004$). This is an artifact of grouping: because CESNET3 contributes no variation in size, the pooled correlation measures network membership under another name. Within the Internet2 group alone, where size varies, the relationship is not significant ($r = -0.43$, $p = 0.216$). The decisive evidence is a matched pair: one backbone router with 1003 windows gains +27.1%, while one campus institution with 1035 loses 4.7%. Data quantity does not order these clients; network membership does.

The interpretation follows from Fig. 2: campus traffic repeats on a stable cycle, so a single institution’s history contains most of what there is to learn, whereas backbone traffic is driven by scheduling external to any router and the shared representation supplies structure local data cannot.

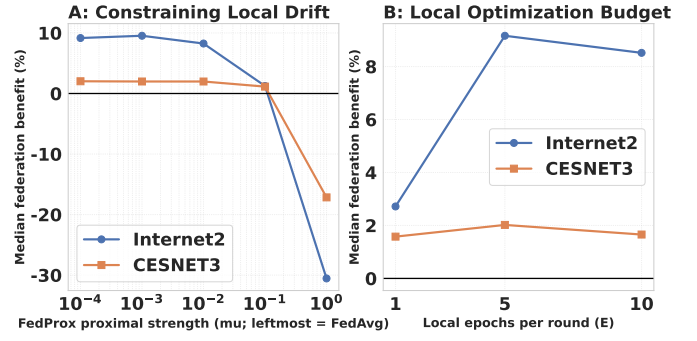


Fig. 5. Two independent manipulations of local optimization. Left: FedProx proximal coefficient. Right: local-epoch budget at fixed communication cost. Both suppress the backbone benefit while leaving the campus benefit unchanged.

D. Local Adaptation Is the Mechanism

If the backbone benefit arises because federation supplies structure absent locally, it should depend on each client’s ability to adapt the shared model during local training, and constraining that adaptation should suppress it selectively. We test this with two independent manipulations (Fig. 5, Table IV).

The first constrains parameter drift directly through the FedProx coefficient of (9). At $\mu \leq 0.01$ the federation is indistinguishable from FedAvg. At $\mu = 0.1$ the backbone benefit collapses to +1.3% while the campus benefit is barely affected, the stratification ceases to be significant ($p = 0.236$), and clients helped fall from 15 to 7. At $\mu = 1$ both groups are harmed.

The second constrains local optimization by budget rather than penalty. Reducing local epochs from $E = 5$ to $E = 1$, with round count and therefore communication cost fixed, reduces backbone benefit from +9.2% to +2.7% while campus benefit moves from +1.9% to +1.6%; raising it to $E = 10$ yields no further gain.

The two manipulations act through different parts of the algorithm yet produce the same selective effect, indicating that the gain for irregular-traffic clients requires substantial unconstrained local adaptation rather than being a property of aggregation alone. This sits athwart the prevailing framing of client drift as a pathology to be corrected [16], [17]: here that divergence is productive for precisely the clients who gain from participating. Recent work reports a similar aggregate trade-off in the local-epoch budget [19]; our results indicate its resolution is client-dependent.

E. Robustness

Table IV collects the robustness checks. Heterogeneity-aware aggregation does not improve on plain averaging here: the best FedProx setting differs from FedAvg by 0.3% against a between-seed deviation of roughly two percent, consistent with evidence that no federated algorithm dominates across non-IID regimes [18].

The stratification is directionally consistent across window lengths and grows with them: the gap between backbone and

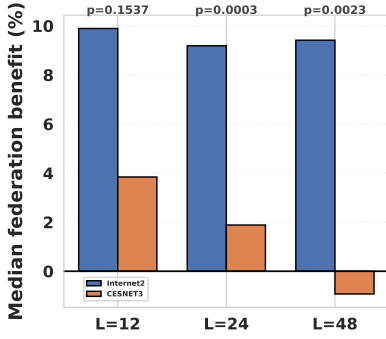


Fig. 6. Median federation benefit by network across window lengths. Direction is consistent and separation grows with window length; the test is not significant at $L = 12$.

campus median benefit is 6.1, 7.3, and 10.3 percentage points at $L = 12$, 24, and 48 (Fig. 6). The test is significant at $L = 24$ and $L = 48$ but not at $L = 12$ ($p = 0.154$), and we scope the claim accordingly. This is what the regime account predicts rather than a failure of it: the distinction between the two regimes is one of diurnal periodicity, and a 12-hour window cannot contain a full daily cycle, so the property separating the groups is only partially observable at that resolution. Consistent with this, we found the $L = 12$ configuration noticeably less stable across repeated runs than $L = 24$ or $L = 48$. Overall benefit nonetheless weakens at $L = 48$ even as the gap widens, with 9 clients helped against 15 at $L = 24$.

F. Secondary Observations

Three further results bear on deployment. First, federation produces a *shared* representation space. Local-only training yields twenty independently initialized models whose latent coordinates are mutually incomparable, whereas federated and centralized training yield a single encoder (Fig. 7). A linear probe recovering network membership from frozen embeddings reflects this ordering (Table III), a prerequisite for any downstream task comparing representations across operators.

Second, federation improves the reliability of model selection. Local-only training attains the *lowest* validation error of the three regimes and the *highest* test error, at a ratio of 3.09 against 1.76 and 1.74 for centralized and federated: each local model selects its stopping point on 95 to 204 windows, whereas the other regimes select against pooled validation data. Federation therefore improves not only what is learned but the reliability with which a deployable checkpoint is chosen, a benefit invisible to validation metrics.

Third, communication cost favors centralization at this data scale: federated averaging transfers roughly 383 MB by (11) against 211 KB to ship every client’s hourly series, so the case for federation rests on the administrative boundary rather than on bandwidth. The ordering inverts at $V^* \approx 19.2$ MB per client by (12), past which sub-hourly, multivariate, or raw-flow deployments sit. Finally, the held-out regional ISP embeds coherently under the federated encoder and falls nearest the

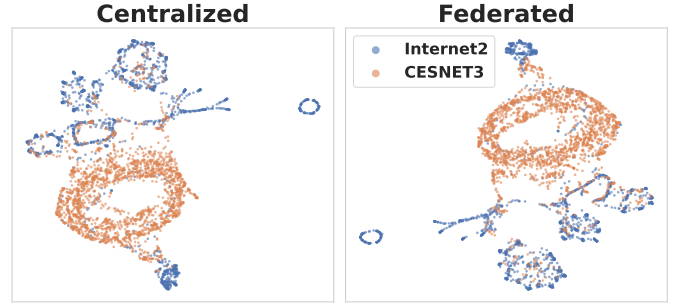


Fig. 7. Projection [32] of test-window embeddings under centralized and federated encoders, colored by network. Both recover the regime distinction with visibly similar structure.

human-driven cluster; with only 25 windows we draw no quantitative conclusion, but the placement is consistent with the regime account above.

VI. DISCUSSION AND LIMITATIONS

A. Implications for Operators and for Federated Optimization

The question an operator faces is not whether federated learning works in general, but whether contributing its own telemetry to a multi-operator federation is worthwhile for that operator specifically. Our results support a concrete answer. A network with stable periodic traffic and a reasonable observation history can expect little; one whose load is driven by external scheduling, or which holds little history, can expect substantially more. This asymmetry appears assessable in advance from properties a prospective participant can inspect without joining, namely the regularity of its own traffic and the length of its own history. It also complicates incentive design, since those with most to gain contribute least.

The mechanism result carries a consequence beyond networking. Client heterogeneity is customarily treated as an obstacle, with methods such as FedProx and SCAFFOLD existing to suppress or correct local drift [16], [17]. In our federation that treatment is counterproductive: heterogeneity here is not noise corrupting a shared signal but part of what makes the shared representation useful, since clients whose local data underdetermines their own behavior benefit from a global model only if permitted to specialize it against their own observations. This does not contradict results motivating drift correction, which concern settings where local optima genuinely conflict, but it suggests that whether drift is pathological is a property of the deployment. A practical corollary is that aggregate metrics can conceal the effect entirely: at $\mu = 0.1$ our federation looked nearly unchanged in mean error while the stratified benefit had become undetectable. A further corollary is that the appropriate degree of personalization is not a single federation-wide hyperparameter but a per-client quantity, which supplies an empirical argument in this domain for personalized federated methods that retain local adaptation by construction [20], [22].

TABLE IV

ROBUSTNESS CHECKS. B IS MEDIAN PER-CLIENT BENEFIT OF (10); W/T/L COUNTS CLIENTS HELPED, UNAFFECTED, AND HARMED AT A TWO-PERCENT THRESHOLD; p IS THE ONE-SIDED MANN-WHITNEY TEST FOR STRATIFICATION. DEFAULT CONFIGURATION IS FEDAVG, $L = 24$, $E = 5$.

Check	Configuration	Fed. MAE	B (Internet2)	B (CESNET3)	W/T/L	p
Aggregation	FedAvg (default)	0.2504	+9.2%	+1.9%	15/4/1	0.0003
	FedProx $\mu = 0.001$	0.2496	+9.8%	+1.8%	15/4/1	0.0002
	FedProx $\mu = 0.01$	0.2512	+8.2%	+1.8%	14/5/1	0.0007
	FedProx $\mu = 0.1$	0.2604	+1.3%	+1.1%	7/12/1	0.236
	FedProx $\mu = 1.0$	0.3411	-30.5%	-17.2%	2/0/18	0.977
Window length	$L = 12$	—	+9.9%	+3.8%	16/3/1	0.154
	$L = 24$ (default)	0.2504	+9.2%	+1.9%	15/4/1	0.0003
	$L = 48$	—	+9.4%	-0.9%	9/7/4	0.002
Local epochs	$E = 1$	0.2574	+2.7%	+1.6%	—	—
	$E = 5$ (default)	0.2504	+9.2%	+1.9%	15/4/1	0.0003
	$E = 10$	0.2522	+8.5%	+1.7%	—	—

B. Limitations

Our federation is simulated: 20 clients execute within one process, exchanging parameters through memory. This is standard for evaluating federated algorithms and appropriate for questions about learning behavior, but it means we measure no communication latency, straggler effects, partial participation, or failure handling. A physically distributed deployment on a programmable testbed such as FABRIC [33] would convert this into a systems contribution. Relatedly, our privacy claim is contractual rather than information-theoretic: raw telemetry need not cross an administrative boundary, but gradient inversion can reconstruct training samples from shared updates alone [23], [24], so a deployment facing an adversarial rather than honest-but-curious server would require differentially private perturbation whose utility cost we do not measure.

The federation is also small, univariate, and selectively sampled: twenty clients is realistic for cross-silo collaboration but far from the scale at which participation strategy becomes consequential, we model a single metric where the CESNET corpus offers twelve per interval, and the CESNET clients are the ten most active of 283 and plausibly the most regular. Our windows span 57 days and nine weeks respectively, which cannot capture annual academic cycles.

Reconstruction error is also a proxy: we do not show that the stratified gain transfers to anomaly detection, forecasting, or any other task an operator performs. The regime account is likewise an interpretation rather than a demonstration. We show the effect is not explained by training-set size and that it tracks the distinction visible in Fig. 2, quantified indirectly through the window-length ablation, but the groups also differ in operator, vantage point, and observation length, and we do not isolate periodicity itself, for instance by synthesizing clients at graded levels of temporal regularity. Such an experiment would test the account directly and is the most valuable single follow-up. Two narrower caveats: our held-out probe comprises 25 windows and supports only a qualitative observation, and our results are specific to a compact autoencoder appropriate to roughly one thousand

windows per client.

VII. CONCLUSION

Operators of research, academic, and commercial networks hold telemetry more valuable modeled jointly than separately, and cannot share it. Evaluating federated representation learning on real data from a national research backbone, a European NREN, and a commercial regional ISP, we found the data-custody constraint is not the obstacle it is often assumed to be: federated training proved statistically equivalent to centralized pooling at a five percent bound. The more consequential finding is that federation does not benefit participants equally. Backbone clients gained a median of nine percent against training alone, while campus institutions gained under two percent, a separation surviving outlier removal, unexplained by data quantity, and tracking a difference in traffic regularity we characterize but do not isolate. Two independent manipulations each selectively removed the benefit for the former group, identifying unconstrained local adaptation as the mechanism and indicating that the heterogeneity federated optimization is generally designed to suppress is productive for exactly the participants with most to gain.

For an operator deciding whether to join, these results make the expected benefit assessable; for the design of federated systems in this domain, they suggest that evaluations reporting only aggregate accuracy may conceal exactly the effects that matter to prospective participants.

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A. Computational Environment

We ran all experiments on one NVIDIA H200 GPU with 16 CPU cores and 60 GB system memory under Python 3.11.13, using PyTorch for models, scikit-learn for the linear probe and scaling, SciPy for statistics, pandas and PyArrow for I/O, and umap-learn for embedding projections. The federation occupies 2.53 MB in single precision and is resident in device memory throughout. End-to-end reproduction of all reported results, including the five-seed main experiment, FedProx sweep, both ablations, and the convergence check, requires approximately three hours of wall-clock time.

B. Data

Internet2 hourly series derive from NetFlow collected over 57 days across ten backbone routers, reduced by the pipeline in [12]; these records are governed by a non-disclosure agreement and cannot be redistributed. CESNET3 series come from the public CESNET-TimeSeries24 corpus [29], specifically the `institutions/agg_1_hour` partition, from which we select the ten institutions with highest mean hourly `n_packets` and truncate each to its first 1512 hours. The regional ISP snapshot comprises 48 hourly aggregates and is not redistributable. Because two of the three sources are restricted, full end-to-end reproduction requires the CESNET corpus together with equivalently formatted hourly series for the remaining clients; all preprocessing, training, and analysis code is released to permit substitution of alternative sources.

C. Reproducibility Protocol

Client construction is deterministic given the source series: chronological partitioning at (0.70, 0.15, 0.15), scaling statistics fitted on the training partition only, and windows extracted independently within each partition at unit stride. Client admission follows the derived threshold of (4) rather than a fixed constant, so eligibility and window configuration cannot diverge.

Every experiment runs under seeds $\{42, \dots, 46\}$ governing initialization, batch ordering, and per-client model initialization under local-only training. Aggregate statistics are the mean across seeds of the per-seed cross-client mean, with dispersion across seeds. Intermediate artifacts are cached at each stage (windowed tensors, checkpoints for all regimes and seeds, training histories, per-client metrics, frozen embeddings), and figure generation reads exclusively from these caches, so all figures and tables regenerate without repeating training. A configuration signature is serialized alongside each artifact and checked on load, so cached results from different settings are detected rather than silently reused. Extending one seed to 200 rounds confirmed convergence at $R = 100$: mean validation error minimized at round 98 and rose slightly thereafter. Model training is seeded but not bit-reproducible across differing GPU hardware, as kernel selection and reduction order vary.

D. Statistical Procedures

Federated against local-only training uses a one-sided Wilcoxon signed-rank test over twenty client-paired errors. Benefit between network groups uses a one-sided Mann-Whitney U test on (10). Equivalence between federated and centralized uses the two one-sided tests procedure [30] with a five percent bound applied to the five paired per-seed means. Reported p -values are one-sided where a directional hypothesis is stated and two-sided otherwise.

E. Artifact Availability

All preprocessing, training, and analysis code, together with the configuration signatures and cached metric tables underlying every figure and table, is available at <https://github.com/mhnafrth/who-benefits-federation>.

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